

Time Complexity
Time Taken to process fetch
,decode and display video

Video Streaming Application

Space Complexity
Memory used for buffering
caching ,Storing and managing
data

Faster Processing
Keeps more data
memory to reduce delay
and Speed

Lower Memory Usage
Uses less memory by reducing cache/ buffer size
or data quality

Tradeoff
Improving one factor may worsen the
other Need to find the optimal balance.

Balanced Resource Utilization
Optimally manages time and memory for smooth
streaming.

Efficient Search Algorithms
Quickly locate and fetch required
video chunks or segments.

Compression Techniques
Reduce Video size Without To
save storage and bandwidth

Buffer Management & Caching
Store frequently used data
in buffer/cache for faster
access.

Algorithm Design

Smooth Video Playback
Ensures uninterrupted and seamless
viewing experience

Reduced Buffering & Lag
Minimizes waiting time and delays during
streaming.

Better User Experience
Higher satisfaction, engagement and retention.

Algorithm Design